TESTING A PROPERTY OF A REPRESENTABLE MORPHISM OF ALGEBRAIC SPACES AFTER AN ETALE COVER

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 10477-10520, immediately following the statement at 10453-10475.

Editorial note. One cross-chapter pointer (Descent) is recorded in assembly.external_references.

Editorial note. The statement quantifies over a property P fixed by the reader; that wording is retained verbatim in the statement span.

Allowed standard prerequisites (editorial).

- Properties of morphisms etale local on the target. the property P fixed in the statement, with Descent, lemma descent-lemma-pullback-property-local-target cited at proof lines 10508-10509. Supplies that P is preserved under etale base change and can be tested after an etale covering of the target.
- Representability of a morphism of algebraic spaces. used throughout: representable morphisms are those whose base change to a scheme is a scheme morphism. Lets the diagrams with V a scheme and V -> Y etale be used as the testing devices.
- Surjectivity and etaleness of the diagonal cover. the morphism U x_X U' -> U' appearing at proof lines 10511-10514. Provides the etale covering of U' which finishes the implication.

Source reference. Original reference 04QU: focused-seeds/geometry/S9-spaces-representable-etale-local-target/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{P}$ be a property of morphisms of schemes which is étale local on the target. Let S be a scheme. Let $f : X \to Y$ be a representable morphism of algebraic spaces over S. Consider commutative diagrams*

$$\begin{array}{ccc} X \times_Y V & \longrightarrow & V \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

where V is a scheme and $V \to Y$ is étale. The following are equivalent

(1) *for any diagram as above the projection $X \times_Y V \to V$ has property $\mathcal{P}$, and*
(2) *for some diagram as above with $V \to Y$ surjective the projection $X \times_Y V \to V$ has property $\mathcal{P}$.*

If X and Y are representable, then this is also equivalent to f (as a morphism of schemes) having property $\mathcal{P}$.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Let us prove the equivalence of (1) and (2). The implication (1) $\Rightarrow$ (2) is immediate. Assume

$$\begin{array}{ccc} X \times_Y V & \longrightarrow & V \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array} \qquad \begin{array}{ccc} X \times_Y V' & \longrightarrow & V' \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

are two diagrams as in the lemma. Assume $V \to Y$ is surjective and $X \times_Y V \to V$ has property $\mathcal{P}$. To show that (2) implies (1) we have to prove that $X \times_Y V' \to V'$ has $\mathcal{P}$. To do this consider the diagram

$$\begin{array}{ccccc} X \times_Y V & \longleftarrow & (X \times_Y V) \times_X (X \times_Y V') & \longrightarrow & X \times_Y V' \\ \downarrow & & \downarrow & & \downarrow \\ V & \longleftarrow & V \times_Y V' & \longrightarrow & V' \end{array}$$

By our assumption that $\mathcal{P}$ is étale local on the source, we see that $\mathcal{P}$ is preserved under étale base change, see Descent, Lemma 04QU. Hence if the left vertical arrow has $\mathcal{P}$ the so does the middle vertical arrow. Since $U \times_X U' \to U'$ is surjective and étale (hence defines an étale covering of U') this implies (as $\mathcal{P}$ is assumed local for the étale topology on the target) that the left vertical arrow has $\mathcal{P}$.

If X and Y are representable, then we can take $\mathrm{id}_Y : Y \to Y$ as our étale covering to see the final statement of the lemma is true. $\square$